DCIT 403 Semester Project Report

University of Ghana

Crop Disease Intelligent Agent

Design and Implementation following Prometheus Methodology

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1. INTRODUCTION

1.1 Problem Statement

Crop diseases are responsible for an estimated 10-16% annual loss in global agricultural production, costing billions of dollars and threatening food security. Smallholder farmers-who produce over 70% of the world's food-are disproportionately affected because they lack timely access to expert plant pathologists. By the time a farmer visually identifies a disease, significant yield damage may have already occurred.

1.2 Why an Agent-Based Approach?

An agent-based approach is ideal for this problem because the system must demonstrate autonomy, reactivity, proactiveness, and goal-directed behaviour-properties that distinguish agents from traditional software applications.

Agent Property	**How It Applies**
Autonomy	The agent independently processes images without human intervention
Reactivity	It reacts to new image inputs (percepts) in real time
Proactiveness	It pursues the goal of accurate diagnosis and can escalate cases
Goal-directed	It has clear goals: accuracy, actionable treatment, and flagging uncertainty

2. SYSTEM ANALYSIS

2.1 Goal Specification

Figure 1: Goal Hierarchy Diagram

3. ARCHITECTURAL DESIGN

3.2 Capability Grouping

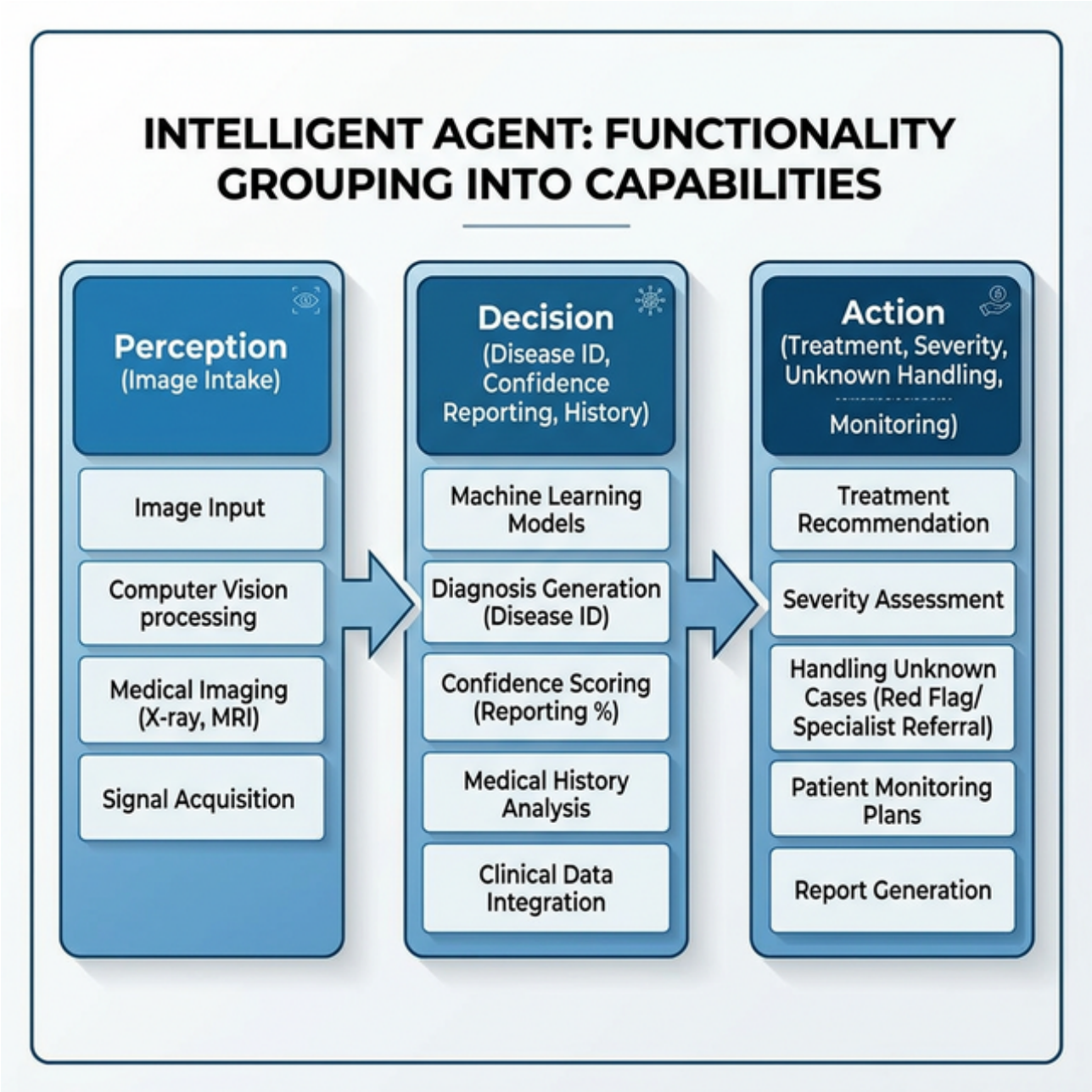

Figure 2: Functionality Grouping Diagram

3.3 Acquaintance Diagram

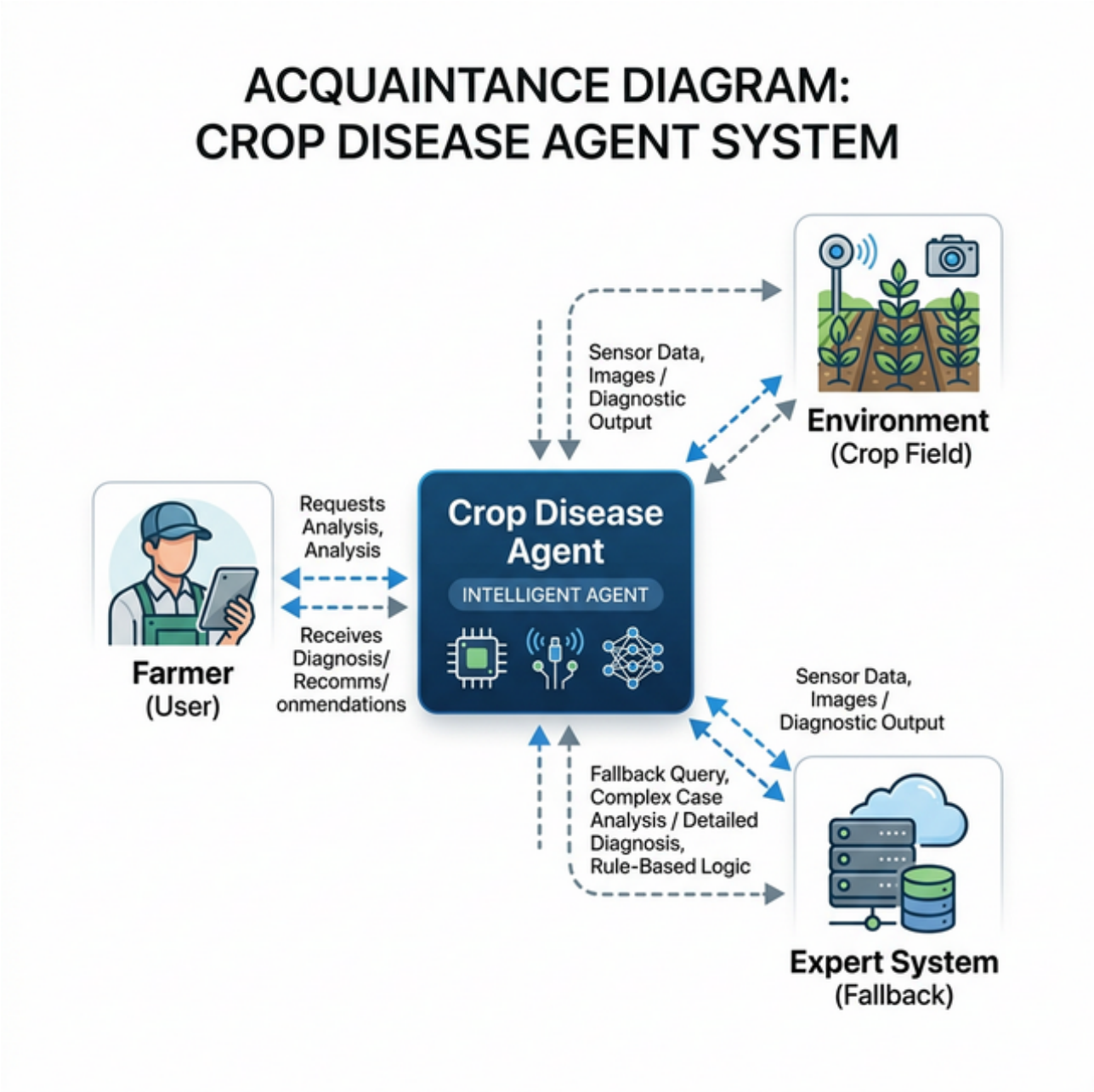

Figure 3: Acquaintance Diagram

4. INTERACTION DESIGN

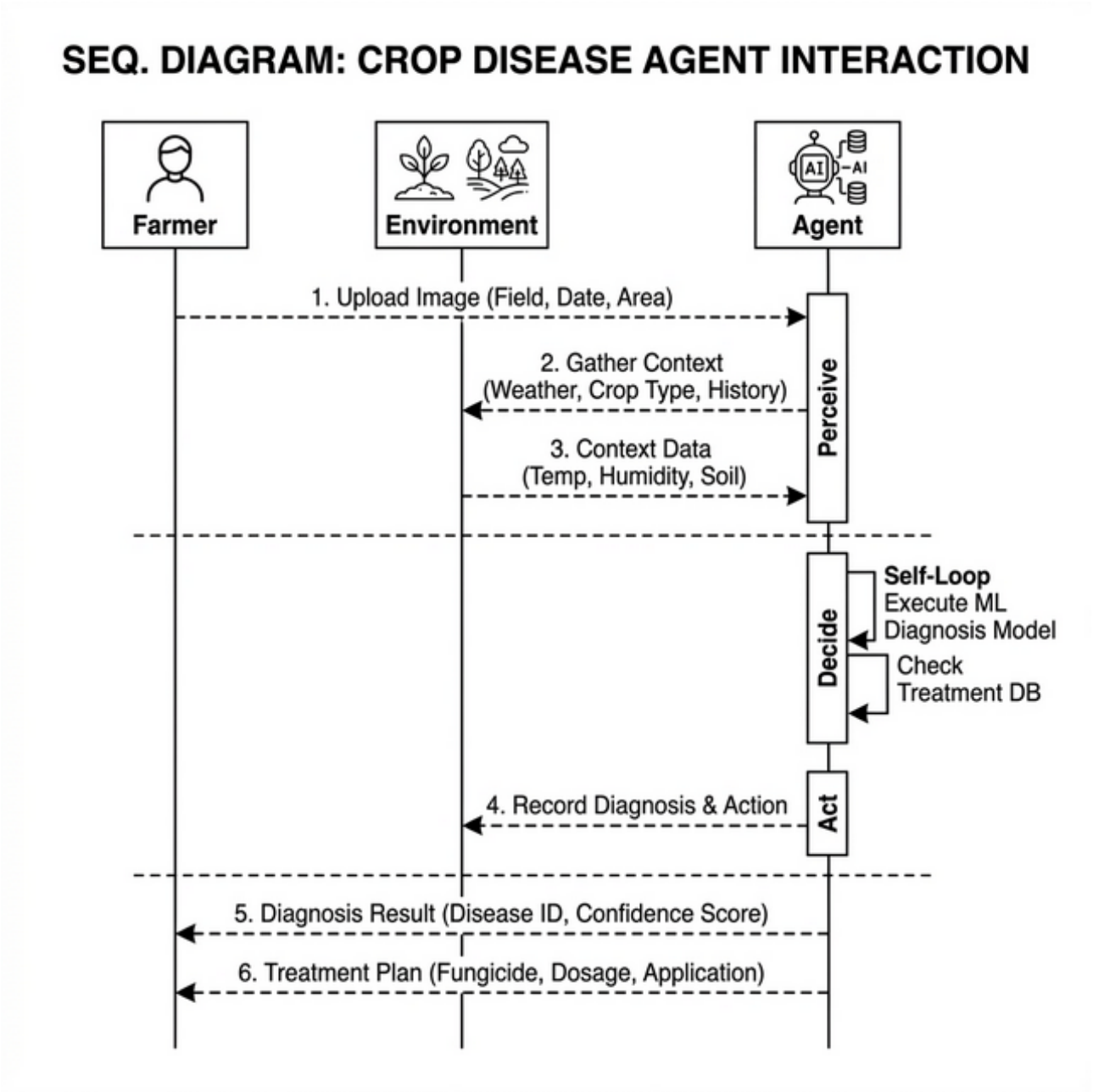

Figure 4: Sequence Interaction Diagram

5. DETAILED DESIGN

The detailed design covers the BDI loop (Perceive-Decide-Act) implemented in Python. Capabilities include Perception, Classification, and Treatment Recommendation.

6. IMPLEMENTATION

The implementation utilizes PyTorch for the CNN pipeline and a

dedicated BDI orchestrator in agent.py. The environment is simulated via environment.py to demonstrate autonomy.

7. CONCLUSION

The Crop Disease Intelligent Agent successfully maps the Prometheus methodology to a real-world domain, providing a robust, autonomous solution for food security.